

Consider two orthogonal matrices U and V of size $n \times n$ then $A = \alpha \begin{bmatrix} aU & bU \\ cV & dV \end{bmatrix}$ will be orthogonal if

- ☐ $\alpha = \frac{1}{\sqrt{2}}, a = 1, b = 1, c = 1, \text{ and } d = -1$
- ☐ $\alpha = \frac{1}{2}, a = 1, b = -1, c = -1, \text{ and } d = -1$
- ☐ $\alpha = \frac{1}{\sqrt{2}}, a = 1, b = -1, c = -1, \text{ and } d = 1$
- ☐ $\alpha = \frac{1}{2}, a = 1, b = 1, c = 1, \text{ and } d = 1$



Question 1 pts



Given matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & \lambda + 1 & \lambda + 1 \\ 0 & 0 & \lambda^2 - 1 \end{bmatrix}$. For which of the values of λ is the rank of A at most 1 ?

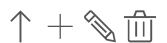
- ☐ 1
- ☐ -1
- ☐ 0
- ☐ 3



Subjective part 1

Group Name

Pick 1 questions, 3 pts per question Pick questions, pts per question



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1.1



Subjective Question 1.1

Consider, for n bigger than 1, an $n \times n$ matrix with all diagonal entries 1, and all off-diagonal entries $\frac{-1}{(n-1)}$. What is the rank of this matrix ?



1.2

**Subjective Question 1.2**

Suppose P, Q are matrices that happen to satisfy $P+Q=PQ$. Then what can be said about the invertibility of $I-P$? Is $PQ=QP$?



1.3

**Subjective Question 1.3**

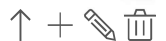
Suppose P is an $n \times m$ matrix.

1. Consider $Q = \begin{pmatrix} P \\ \sqrt{\lambda}I \end{pmatrix}$, for some $\lambda > 0$. Is Q onto? [1 point]

2. Consider $R = P^T P + \lambda I$, for some $\lambda > 0$. Is R one-one? Is R onto? [2 points]

**Subjective part 2**

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2.1

**Subjective Question 2.1**

Suppose A and B are 4×2 and 2×4 matrices respectively, such that $AB = 0$ (i.e. AB is the all-zero matrix). What are the possible values of $\text{rank}(A)$ and $\text{rank}(B)$? For each possible value of the ranks, construct a suitable example of A and B with those ranks such that $AB = 0$. If certain combinations of rank are impossible, clearly justify.

General answer comments

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2.3

**Subjective Question 2.3**

Given 2x2 matrices A , B , is it true that $\text{rank}(AB) = \text{rank}(BA)$?



2.2



Subjective Question 2.2

Suppose A and B are 2x4 and 4x2 matrices respectively, such that $AB = I$. What are the possible values of $\text{rank}(A)$ and $\text{rank}(B)$? For each possible value of the ranks, construct a suitable example of A and B with those ranks such that $AB = I$. If certain combinations of rank are impossible, clearly justify.

General answer comments

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Subjective part 3

Pick 1 questions, 5 pts per question Pick questions, pts per question





3.1



Subjective Question 3.1

In this problem, we are investigating 2x2 symmetric orthogonal matrices.

1. Does any 2x2 symmetric orthogonal matrix have to be a diagonal matrix? [2 points]

2. Explain if the following statement is true, with justification. [3 points]

"Any 2x2 symmetric orthogonal matrix A operates by *reflection*: Any input x is reflected about some line to result in Ax ."

If true, identify the line about which the input x is reflected and give proof. If false, provide a counter-example.

General answer comments

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3.2



Subjective Question 3.2

Consider n users who participate in a social network, and the corresponding $n \times n$ *friendship matrix* A whose entries are

$$A_{ij} = \begin{cases} 1 & \text{if users } i, j \text{ are friends,} \\ 0 & \text{otherwise.} \end{cases}$$

We define the diagonal entries of this matrix to be zero.

1. Let B be an $n \times n$ matrix, whose entry B_{ij} is equal to the number of friends users i and j have in common. The diagonal entry B_{ii} of B is taken to be the number of friends of user i . Give a "simple" relationship between the matrices A and B . [2 points]

2. If the largest number of friends any user has is k , then show that

$$\underline{x}^T A \underline{x} \leq k \underline{x}^T \underline{x} \text{ for any vector } \underline{x} \in \mathbb{R}^n. [2 \text{ points}]$$

3. Suppose $n=3$. What are the possible ranks of the matrix A ? [1 point]



3.3



Subjective Question 3.3

We say that a vector $\underline{x} \in \mathbb{R}^n$ is a *probability vector* if all the entries in $\underline{x}$ are non-negative (i.e. $x_i \geq 0$ for all i) and the entries sum to 1 (i.e. $x_1 + x_2 + \dots + x_n = 1$). Let $\mathcal{P}_n$ denote the set of all probability vectors in $\mathbb{R}^n$.

1. Sketch $\mathcal{P}_2$. In general, is $\mathcal{P}_n$ a subspace of $\mathbb{R}^n$? [1 point]

Now, we are interested in $n \times n$ matrices P such that

$$P\underline{x} \text{ is a probability vector whenever } \underline{x} \text{ is.} \quad (1)$$

2. Show that P satisfies the statement (1) above *if and only if* all the columns of P are probability vectors. [3 points]

3. Suppose you are given a matrix P and asked to *verify* if it satisfies the statement (1) above. How many computations (additions and multiplications) are required to complete this verification? [1 point]



Subjective part 4

Pick 1 questions, 6 pts per question Pick



questions,



pts per question



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4

**Subjective Question 4**

Consider matrices A , B , and their product AB . Show the following:

1. The null space of B is contained in the null space of AB . [1 point]

So if the nullity of B is k and the nullity of AB is s , then $s \geq k$. The above statement allows us to do the following: We let $\underline{v}_1, \underline{v}_2, \dots, \underline{v}_k$ be a basis for the null space of B , then we can expand it to a basis $\underline{v}_1, \underline{v}_2, \dots, \underline{v}_k, \underline{v}_{k+1}, \dots, \underline{v}_s$ for the null space of AB (you do not have to prove this for the exam. This follows from the way we defined basis in class).

2. The vectors $B\underline{v}_{k+1}, B\underline{v}_{k+2}, \dots, B\underline{v}_s$ are linearly independent. [2 points]

3. The nullity of A is at least $s - k$. [1.5 points]

4. $\text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - n$. Here n is the number of columns in A . [1.5 points]

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